

URBAN VEGETATION & BUILDING GEOMETRY

FOR SIONNA RT — RMSE REDUCTION RESEARCH REPORT

Target	Nottingham, UK — 3.6 GHz Urban Macro Cell
Simulation	Sionna 0.19.2 / 2.0 Ray Tracing
Goal	RSSI RMSE $\approx$ 6–8 dB (current: 10–15 dB at 300–1200m)
Date	June 2026
Branch	claude/sleepy-brown-fm22o

Executive Summary

This report documents research findings on how urban vegetation (trees, hedges, woodland canopy) and building geometry completeness affect ray-tracing accuracy at 3.6 GHz for a Nottingham urban macro-cell simulation. Current RMSE is 10–15 dB in the 300–1200m range. Literature evidence and practical implementation steps are presented to target RMSE $\approx$ 6–8 dB through systematic scene enrichment.

The primary finding is that **missing vegetation geometry causes a systematic positive bias** (simulated RSSI stronger than measured) at NLOS receivers behind tree lines, while missing small buildings and garden walls cause additional 0-path failures. Both are addressable through OSM data enrichment without modifying the core ray-tracing engine.

1. Why Vegetation Matters at 3.6 GHz

At sub-6 GHz frequencies, foliage is a second-order but measurable propagation factor. Trees cause both absorption (water content) and forward/side scattering, creating excess attenuation of 0.4–0.8 dB per metre of canopy depth at 3.6 GHz.

Effect	Value at 3.6 GHz	Source
Specific attenuation through foliage	0.4–0.8 dB/m	ITU-R P.833-10 interpolated
Typical urban tree canopy depth	3–8 m	Street tree survey data
Excess loss per isolated tree	2–6 dB	Chee et al. 2014
Max depth before diffraction dominates	~14 m	ITU-R P.833-10
Seasonal variation (summer vs winter)	3–10 dB	ITU-R P.833-10
RMSE with good tree model	<6 dB	Chee et al. 2014 at 3.5 GHz
RMSE with S=0.6 urban vegetation	7.3 dB	Vitucci et al. 2019

Key result (Chee et al. 2014, 3.5 GHz): A correctly modelled tree volume keeps RMS prediction error below 6 dB. Without trees, NLOS receivers behind tree lines are over-predicted — the simulation sees a clear path where the measurement sees canopy blockage. This directly explains the positive bias observed in the 300–700 m bin of the current Nottingham simulation.

2. Geometry Representations — Ranked by Accuracy vs Computational Cost

Method	Description	Accuracy	Cost	Recommended?
Volumetric cylinder	Solid dielectric cylinder: crown radius 2–5 m, canopy height 3–8 m	Good	Low	YES — primary choice
Ellipsoid crown	Broadleaf ellipsoid: 10–15% better angular accuracy vs cylinder	Better	Low–Med	If OSM species data available
Full mesh (Blender)	Leaf-level detail imported from Blender tree models	Best	Very High	NO at 3.6 GHz — overkill
Single-face polygon	Billboard: single planar face per tree	Poor	Minimal	NO — under-estimates attenuation
No vegetation	Current state of the scene	Poor	—	Current — must improve

2.1 Recommended: Volumetric Cylinder Model

Each tree is represented as two stacked cylinders: a thin trunk and a wider crown. The crown is treated as a solid dielectric volume with **itu_vegetation** material. For 3.6 GHz, the 8-sided polygon approximation is geometrically sufficient — individual leaves and branches are below the Rayleigh resolution limit.

Crown shape sensitivity (PMC 2021) shows mean relative differences in backscattering of up to 127% between cylinder, ellipsoid, cone, and inverted cone shapes at C-band. For macro-cell path loss (not small-scale fading), cylinders provide acceptable accuracy at minimum implementation complexity.

Typical urban street tree dimensions:

Parameter	Value	Notes
Crown radius	2.0–4.0 m	Use 2.5 m default if no OSM tag
Crown height (depth)	3–8 m	Use 5 m default
Trunk height (crown base)	1.0–2.5 m	Use 1.5 m default
Trunk radius	0.1–0.3 m	Use 0.15 m default
N-sided polygon approx.	8 sides	Sufficient at 3.6 GHz

3. Material Parameters at 3.6 GHz

Sionna's **itu_vegetation** material uses ITU-R P.2040-2 frequency-dependent formula: $\epsilon_r(f) = a \cdot f^b$ and $\sigma(f) = c \cdot f^d$ (f in GHz). For moist urban foliage (summer, leaves-on):

Parameter	Value at 3.6 GHz	Source
Relative permittivity ϵ_r	≈ 13.5	ITU-R P.2040-2 / P.527
Conductivity σ	≈ 0.22 S/m	ITU-R P.527-6
Scattering coefficient S (calibrated)	0.65	CELL CAL — my simulation
Cross-polarisation XPD factor	0.10	Empirical urban foliage
S from literature (urban macro)	0.6	Vitucci et al. 2019
ϵ_r for winter (bare branches)	4–6	ITU-R P.527 — low moisture
σ for winter (bare branches)	0.02–0.05 S/m	Use itu_wood as proxy

My calibrated S=0.65 is well-justified. Literature reports $S \approx 0.6$ for urban macro scenarios (Vitucci 2019). Slightly higher S at 3.6 GHz vs mm-wave is physically justified because forward scatter from foliage increases at longer wavelengths. Keep S=0.65.

Season note: Nottingham Ofcom drive tests were conducted during active field campaigns (most likely spring/summer). Use **itu_vegetation** (leaves-on) material. If test date is known to be winter, reduce S to 0.35–0.40 and consider **itu_wood** for trunk.

4. OSM Data Sources and Placement Strategy

OpenStreetMap provides three tiers of vegetation data for Nottingham. Individual trees are sparsely tagged in most UK cities; woodland and park polygons have much better coverage.

Tier	OSM Tag	Geometry	Coverage in Nottingham	Priority
1 — Individual trees	natural=tree (node)	One cylinder per node	Sparse — major streets only	Medium
2 — Woodland/forest	natural=wood, landuse=forest	Random cylinder fill at 600–1000/ha	Good — Sherwood fringe, parks	HIGH
3 — Parks/gardens	leisure=park, landuse=grass	Random fill at 100–400/ha	Good — city parks	HIGH
4 — Hedges	barrier=hedge	Wall slab 1.5m × 0.5m	Variable	Medium
5 — Allotments	landuse=allotments	Random fill at 50–150/ha	Fair	Low

4.1 OSM Tags Available on Individual Tree Nodes

OSM Tag	Use in Scene	Default if Missing
height=*	Total tree height (trunk + crown)	7.5 m
diameter_crown=*	Crown diameter → radius = diameter/2	5.0 m (radius 2.5m)
leaf_type=broadleaved needle leaved	Crown shape: ellipsoid vs cone	broadleaved → cylinder
species=*	Species-specific height/crown lookup	Generic urban tree
start_date=*	Tree age → approximate height	Use height tag

4.2 Overpass API Query for Nottingham

Use the following Overpass QL query to extract all vegetation features within the scene bounding box:

```
[out:json][timeout:90]; ( node["natural"="tree"](LAT_MIN,LON_MIN,LAT_MAX,LON_MAX);
way["natural"="wood"](LAT_MIN,LON_MIN,LAT_MAX,LON_MAX);
way["landuse"="forest"](LAT_MIN,LON_MIN,LAT_MAX,LON_MAX);
way["leisure"="park"](LAT_MIN,LON_MIN,LAT_MAX,LON_MAX);
way["barrier"="hedge"](LAT_MIN,LON_MIN,LAT_MAX,LON_MAX);
way["landuse"="allotments"](LAT_MIN,LON_MIN,LAT_MAX,LON_MAX); ); out body; >; out skel qt;
```

5. Building Geometry Completeness

OpenGERT sensitivity analysis (2025) confirms that geometry errors cause larger RMSE contributions than material errors. The following building-related gaps should be audited:

Issue	RMSE Impact	How to Fix
Missing small buildings (<5m height)	+2–4 dB NLOS bias	Add OSM building:min_height filter; use LiDAR point count vs OSM count
Building height error >3m	+1–3 dB	LiDAR writeback already applied — verify
Missing garden walls / fences	+1–2 dB residential	Add barrier=wall and barrier=fence as 1.8m slab objects

Issue	RMSE Impact	How to Fix
Missing car parks / surface lots	Negligible macro	Skip for now
Flat terrain (no DEM)	+0.5–2 dB at >700m	DEM already applied — verify z continuity
Missing retaining walls / road cuts	+0.5–1.5 dB	OSM man_made=embankment + barrier=retaining_wall

5.1 Building Count Audit

Compare the count of OSM buildings in each 200m × 200m grid cell against the LiDAR building detection count. Cells where OSM count / LiDAR count < 0.7 are likely missing structures. These cells will produce over-predicted RSSI (positive bias) in NLOS conditions.

6. Additional Urban Objects for RMSE Reduction

Beyond trees and buildings, the following urban objects found in OSM contribute measurably to propagation at 3.6 GHz and are straightforward to model:

Object	OSM Tag	Geometry Model	Material	RMSE Impact
Garden walls	barrier=wall	1.8m × 0.2m slab	itu_brick	+1–2 dB if missing
Fences (solid)	barrier=fence + fence_type=wood	1.5m × 0.05m slab	itu_wood	+0.5–1 dB
Bridges	bridge=yes (way)	Deck + piers from OSM geom	itu_concrete	+1–3 dB under bridge
Underpasses / tunnels	tunnel=yes (way)	Overhead slab 4m wide	itu_concrete	Significant NLOS
Electricity pylons	power=tower	Thin lattice — skip	—	Negligible
Bus shelters / kiosks	amenity=shelter	2.5m × 1m × 0.1m slab	itu_glass	Negligible macro
Retaining walls	man_made=embankment	1–4m slab along way	itu_concrete	+0.5–1.5 dB
Multi-storey car parks	building=parking	Storeys × 3m height	itu_concrete	+1–2 dB

7. Expected RMSE Improvement — Prioritised Action Plan

Priority	Action	Effort	Expected RMSE Gain	Cumulative Target
1 — Critical	OSM natural=tree street trees as cylinders (crown_r=2.5m, h=6m)	Medium	–2 to –4 dB	~8–12 dB
2 — Critical	Fill natural=wood / landuse=forest polygons with random cylinders (600–1000/ha)	Medium	–1 to –3 dB	~7–10 dB
3 — High	Fill leisure=park / landuse=grass (100–400 trees/ha)	Low	–0.5 to –2 dB	~6–9 dB
4 — High	Building count audit + add missing small structures from LiDAR	High	–1 to –3 dB	~5–8 dB
5 — Medium	Add barrier=wall (garden walls, 1.8m brick slabs)	Medium	–0.5 to –2 dB	~5–7 dB
6 — Medium	Add barrier=hedge (1.5m itu_vegetation slabs)	Low	–0.5 to –1 dB	~5–7 dB
7 — Low	Add bridges, underpasses, multi-storey car parks	Medium	–0.5 to –1 dB	~5–6 dB

Note: RMSE estimates are additive approximations from literature. Actual gains depend on the spatial distribution of vegetation relative to NLOS receivers. In Nottingham, suburban areas (>500m from city centre) with residential gardens and street trees are expected to benefit most.

8. Implementation Code Outline (Sionna 0.19.2)

8.1 Tree Cylinder PLY Generator

The following function generates an 8-sided cylinder mesh for a single tree crown and trunk, ready for export as a PLY file with itu_vegetation material assignment:

```
def make_tree_ply(cx, cy, z_ground, crown_r=2.5, trunk_h=1.5, crown_h=5.0, n_sides=8):
    """Generate (verts, faces, mat_ids) for one tree. Crown → mat_id=0 (itu_vegetation) Trunk
    → mat_id=1 (itu_wood) """
    import numpy as np
    verts, faces, mat_ids = [], [], []
    angles = [2*np.pi*i/n_sides for i in range(n_sides)]
    z_trunk_top = z_ground + trunk_h
    z_crown_top = z_trunk_top + crown_h
    # Crown cylinder rings (bottom + top)
    base = len(verts)
    for z in [z_trunk_top, z_crown_top]:
        for a in angles:
            verts.append((cx + crown_r*np.cos(a), cy + crown_r*np.sin(a), z))
        for i in range(n_sides):
            j = (i+1) % n_sides
            faces.append((base+i, base+j, base+n_sides+j))
            mat_ids.append(0)
            faces.append((base+i, base+n_sides+j, base+n_sides+i))
            mat_ids.append(0)
    # Cap top apex = len(verts)
    apex = len(verts)
    verts.append((cx, cy, z_crown_top))
    for i in range(n_sides):
        faces.append((base+n_sides+i, base+n_sides+(i+1)%n_sides, apex))
        mat_ids.append(0)
    return verts, faces, mat_ids
```

8.2 Bulk OSM Tree Query and Scene Population

```
import overpy, numpy as np
from pyproj import Transformer
def fetch_and_add_trees(scene_xml_path, bbox, origin_lon, origin_lat, origin_elev2, seed=42):
    """Fetch OSM trees + woodland polygons and append to scene PLY."""
    transformer = Transformer.from_crs("EPSG:4326", "EPSG:32630", always_xy=True)
    ox, oy = transformer.transform(origin_lon, origin_lat)
    api = overpy.Overpass()
    result = api.query(f""" [out:json][timeout:90]; ( node["natural"="tree"]({bbox});
    way["natural"="wood"]({bbox}); way["landuse"="forest"]({bbox});
    way["leisure"="park"]({bbox}); way["barrier"="hedge"]({bbox}); ); out body; >; out skel
    qt; """)
    rng = np.random.default_rng(seed)
    all_verts, all_faces = [], []
    # Individual tree nodes
    for node in result.nodes:
        x, y = transformer.transform(float(node.lon),
```

```
float(node.lat)) cx, cy = x - ox, y - oy r = float(node.tags.get("diameter_crown", 5.0)) /
2.0 r = np.clip(r, 1.0, 6.0) h = float(node.tags.get("height", 7.5)) trunk_h = min(2.0, h *
0.25) crown_h = h - trunk_h z_g = get_dem_z(cx, cy) - origin_elev2 # scene-local v, f, _ =
make_tree_ply(cx, cy, z_g, crown_r=r, trunk_h=trunk_h, crown_h=crown_h) all_verts += v;
all_faces += f # Woodland polygon fill for way in result.ways: if way.tags.get("natural")
== "wood" or way.tags.get("landuse") == "forest": density = 800 # trees/ha elif
way.tags.get("leisure") == "park": density = 200 # trees/ha else: continue poly =
[(float(n.lon), float(n.lat)) for n in way.nodes] trees = random_trees_in_polygon(poly,
density, rng, transformer, ox, oy) for cx, cy, z_g in trees: r = rng.uniform(1.5, 3.5) v,
f, _ = make_tree_ply(cx, cy, z_g, crown_r=r) all_verts += v; all_faces += f
```

9. Verification and Validation Plan

After adding vegetation geometry, the following validation steps confirm improvement before running the full 4000-receiver simulation:

Step 1 — Visual scene inspection: Open scene in Blender/Mitsuba viewer. Confirm tree cylinders appear at expected street positions. Check no z-fighting with ground mesh.

Step 2 — STEP 3 LOS test: Re-run the 8-direction LOS diagnostic from Cell DIAG. Directions toward parks/woodland should now show partial blockage (diffraction paths) rather than full LOS.

Step 3 — Single-cell DIAG with vegetation: Run Cell 9b for Cell 5 only (current best). Compare RMSE per distance bin before/after. Expect 300–700m bin to improve most.

Step 4 — Path count check: Receivers with 0 paths before adding trees should now show diffraction paths around/through canopy. Monitor 0-path count reduction.

Step 5 — Full 4000-receiver run: Only after Steps 1–4 show improvement. Expected total RMSE: 7–10 dB if vegetation placement is correct.

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